

EEEM078 – Computer Vision and Pattern Recognition

Content-Based Image Retrieval using Colour Histograms and Spatial Descriptors

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Abstract

This coursework investigates the effectiveness of different feature representations for content-based image retrieval using the MSRC image dataset. The retrieval system follows a query-by-example framework in which images are represented by feature descriptors and ranked according to similarity to a query image.

A baseline retrieval system based on a global RGB colour histogram was first implemented. The histogram descriptor captures the overall colour distribution of an image but does not incorporate spatial information. Evaluation using Mean Average Precision (mAP) showed that this approach achieved moderate performance, with an mAP of 0.277.

To improve retrieval accuracy, a spatial colour histogram descriptor was introduced by dividing images into a 2×2 grid and computing histograms for each region. This spatial representation improved performance to an mAP of 0.292 by incorporating coarse structural information into the feature representation.

Finally, Principal Component Analysis (PCA) was applied to reduce feature dimensionality while preserving 95% of the variance. Although PCA reduced the feature dimension from 2048 to 137, retrieval performance decreased to an mAP of 0.244, highlighting the trade-off between dimensionality reduction and discriminative power.

Overall, the experiments demonstrate the importance of spatial information in colour-based retrieval systems and illustrate the limitations of purely colour-based descriptors.

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1. Dataset and Experimental Setup

1.1 Dataset

The experiments were conducted using the MSRC Object Category Image Database (Microsoft, 2024), which contains 4323 images across 18 semantic categories, each corresponding to a top-level directory.

The categories include: aeroplanes, animals, benches_and_chairs, bicycles, birds, buildings, cars, chimneys, clouds, doors, flowers, kitchen_utensils, leaves, miscellaneous, scenes, signs, trees and windows.

Each image is labelled according to its top-level folder, which defines the ground-truth category. An image is considered relevant if it belongs to the same category as the query.

The dataset exhibits significant variation in object appearance, background, and lighting conditions. While some categories (e.g., cars, windows, clouds) are visually consistent, others (e.g., miscellaneous, buildings, signs) are more heterogeneous, making the dataset suitable for evaluating retrieval robustness.

1.2 Pre-processing

All images were resized to a fixed resolution of 256×256 pixels prior to feature extraction in order to ensure consistent feature dimensionality and computational efficiency. Images were converted to the RGB colour space, which was used to compute colour-based descriptors.

No additional segmentation or object localisation was performed; descriptors were extracted from the entire image. As a result, global image appearance and background information influence the feature representation.

1.3 Retrieval Framework

The retrieval system follows a query-by-example paradigm. For a given query image, a feature vector is extracted and compared with the feature vectors of all other images in the dataset. Images are ranked according to their similarity to the query.

Several feature representations were evaluated in this coursework:

1. Global RGB colour histogram (baseline descriptor)
2. Spatial RGB histogram using a 2×2 grid
3. Spatial histogram with dimensionality reduction using PCA

The similarity between images was computed using different distance metrics depending on the feature representation:

- Euclidean distance
- Cosine distance
- Chi-square distance (particularly suitable for histogram-based descriptors)

1.4 Evaluation Protocol

Retrieval performance was evaluated using Mean Average Precision (mAP) across all query images. For each query, the ranked retrieval list was compared with the ground-truth relevant set (images sharing the same category label).

Additional evaluation measures were used to analyse retrieval behaviour:

- Per-class Average Precision (AP)
- Precision–Recall curves
- Confusion matrices based on top-10 retrieval voting
- Analysis of the most frequent class confusions

All images in the dataset were used as query images to ensure a comprehensive evaluation of the system.

2. Baseline Image Retrieval Using Global Colour Histograms

2.1 Feature Representation

The baseline image representation used in this study is a global RGB colour histogram. Colour histograms provide a compact description of the distribution of colours within an image and are widely used in content-based image retrieval due to their simplicity and robustness to small geometric variations (Swain and Ballard, 1991).

For each image, a three-dimensional histogram was computed over the RGB colour space. Each colour channel was quantised into 8 bins, resulting in a total histogram dimensionality of:

$$8 \times 8 \times 8 = 512$$

Each image was therefore represented by a 512-dimensional feature vector describing the global distribution of colours across the entire image. After computation, each histogram was L1-normalised so that the sum of all bins equals one. This normalisation ensures that feature vectors are comparable regardless of image brightness or size.

The global colour histogram does not incorporate spatial information; it only describes the overall colour distribution. Consequently, images with similar colours but different spatial arrangements may still appear similar in the feature space.

2.2 Similarity Measures

To compute similarity between images, several distance metrics were evaluated:

- Euclidean distance
- Cosine distance
- Chi-square distance

Among these metrics, the Chi-square distance is particularly well suited for histogram-based descriptors because it compares bin-wise differences between histograms and emphasises differences in small values (Szeliski, 2022).

For a given query image, the distance between the query feature vector and all other image feature vectors in the dataset was computed. Images were then ranked according to increasing distance, producing a ranked retrieval list.

2.3 Baseline Retrieval Behaviour

Figure 1 illustrates an example retrieval result using the global RGB histogram descriptor. The query image belongs to the *cars* category, and the system retrieves visually similar images ranked by histogram similarity.

Several retrieved images correspond correctly to the *cars* category; however, some incorrect results also appear in the top-ranked positions. For example, bicycles, scenes, signs or miscellaneous may occasionally be retrieved due to similarities in global colour distribution. This behaviour occurs because the descriptor captures overall colour composition but does not encode object shape or spatial layout.

This limitation demonstrates that global colour histograms can produce reasonable retrieval results but may struggle to distinguish visually different objects that share similar colour characteristics.



Figure 1: Example retrieval result using the global RGB histogram descriptor.

2.4 Baseline Performance

The baseline retrieval system was evaluated using Mean Average Precision (mAP) across all query images. Using the chi-square distance metric, the baseline system achieved:

$$\text{mAP} = 0.277$$

This result indicates moderate retrieval performance. Categories with distinctive colour distributions, such as *clouds* and *cars*, generally achieve higher precision, whereas visually heterogeneous categories such as *miscellaneous* or *buildings* tend to produce lower retrieval accuracy.

The baseline results provide a reference point for evaluating more advanced descriptors introduced in later sections.

3. Retrieval Evaluation and Analysis

3.1 Evaluation Metrics

Retrieval performance was evaluated using Mean Average Precision (mAP) (Bishop, 2006). For each query, images from the same category were treated as relevant, and Average Precision (AP) was computed based on their ranking. The mAP was then calculated as the mean AP across all queries.

Additional evaluation included:

- Precision–Recall curves to analyse performance across recall levels
- Per-class Average Precision to assess category-wise performance
- Confusion matrices to identify systematic errors

These measures provide a comprehensive evaluation of the retrieval system.

3.2 Precision–Recall Analysis

Figure 2 shows the precision–recall curve for the baseline descriptor. Precision is high at low recall because the most similar images are retrieved first. As recall increases, precision decreases as less relevant images are included.

This behaviour reflects a limitation of global descriptors, which capture overall appearance but lack structural detail.

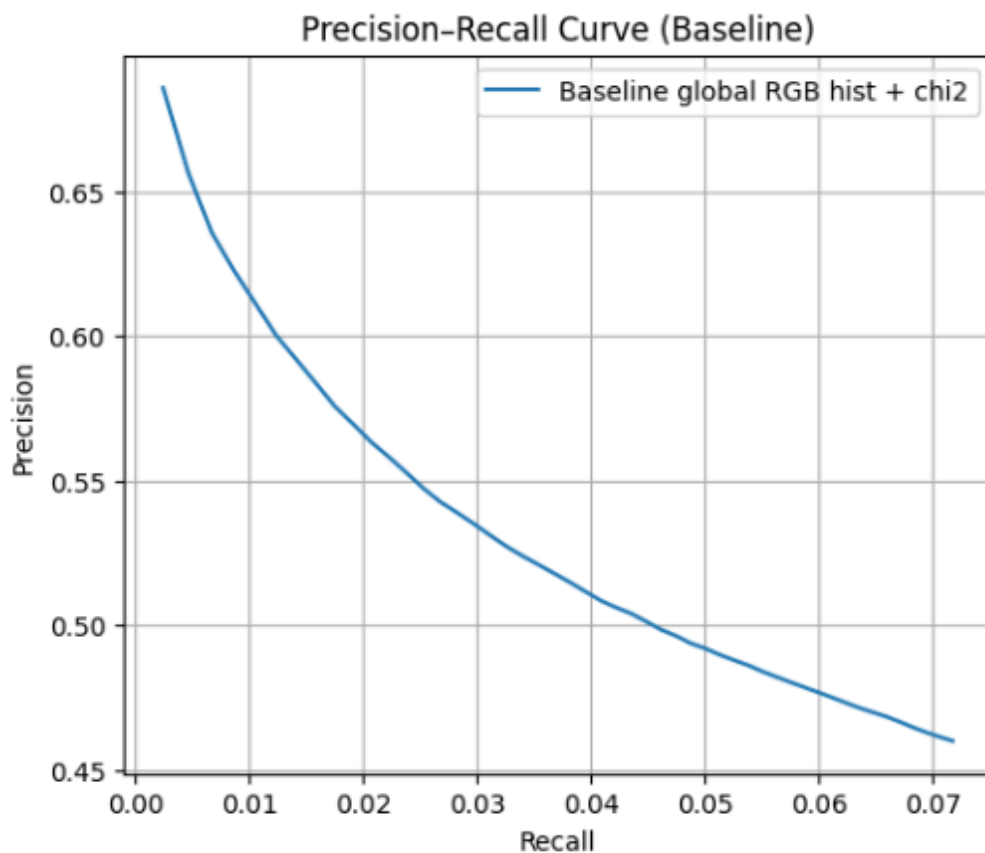


Figure 2: Precision–Recall curve for the baseline RGB histogram retrieval system using the chi-square distance metric.

3.3 Class-Level Performance

Per-class analysis shows that some categories achieve higher performance than others. The best-performing categories include clouds, cars, chimneys, windows and animals, which tend to have distinctive colour patterns or consistent structure.

In contrast, categories such as signs, benches_and_chairs, birds, miscellaneous and buildings perform poorly due to greater visual variability or small object size. In these cases, the descriptor is often dominated by background colours.

Overall mAP (chi2): 0.2773786065668997	
Top 5 classes by AP:	
clouds	0.559
cars	0.421
chimneys	0.371
windows	0.355
animals	0.312
Bottom 5 classes by AP:	
signs	0.041
benches_and_chairs	0.045
birds	0.048
miscellaneous	0.052
buildings	0.067

Figure 3: Class-level retrieval performance for the baseline RGB histogram descriptor using the chi-square distance metric.

3.4 Confusion Analysis

Figure 4 presents the confusion matrix based on top-10 retrieval results. Each row represents the true class, while each column represents the predicted class.

The matrix reveals systematic errors. For example, doors are frequently confused with windows due to similar visual characteristics. Other confusions include buildings with windows, benches_and_chairs with bicycles, and scenes with trees.

These results highlight a key limitation of global colour histograms: they ignore spatial structure, making it difficult to distinguish categories with similar colour distributions.

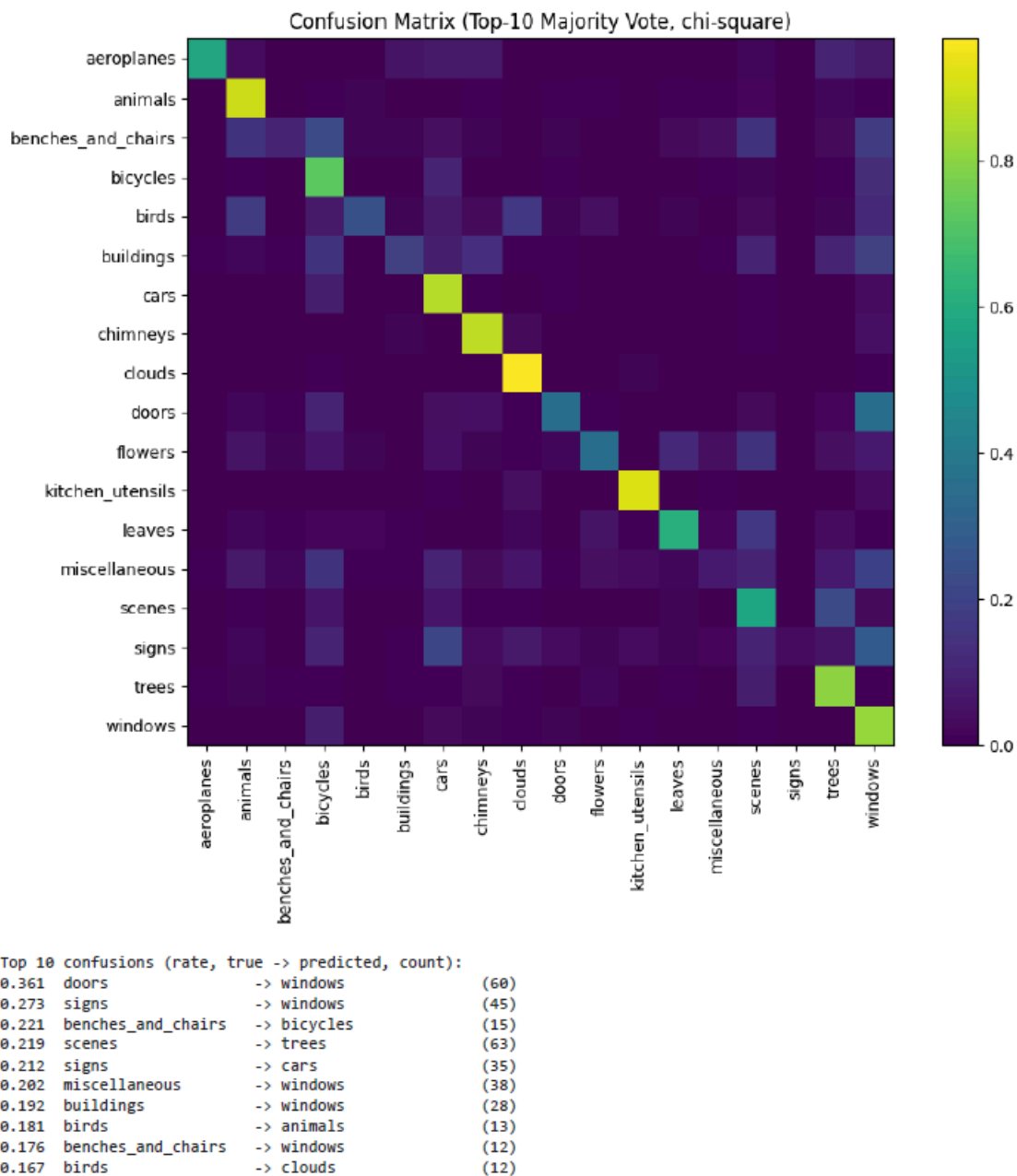


Figure 4: Confusion matrix for the baseline image retrieval system using the global RGB histogram and chi-square distance.

3.5 Summary of Baseline Limitations

The results show that the global colour histogram is a simple but limited representation. While it performs well for visually distinctive categories, it struggles when spatial structure is important.

These limitations motivate the use of spatial descriptors, which incorporate structural information to improve retrieval performance.

4. Spatial Colour Histogram Descriptor

4.1 Motivation

The baseline global colour histogram revealed clear limitations. While it captures overall colour distribution, it does not encode spatial arrangement. As a result, images with similar colours but different structures may appear similar in the feature space.

This is reflected in frequent confusions between categories such as doors and windows or buildings and windows, where colour composition is similar but spatial layout differs. These observations suggest that incorporating spatial information can improve retrieval performance (Szeliski, 2022).

4.2 Spatial Histogram Representation

To address this limitation, a spatial RGB histogram was implemented by dividing each image into a 2×2 grid, producing four regions.

An RGB histogram was computed for each region using the same bin configuration as the baseline (8 bins per channel), resulting in 512 bins per region. The four histograms were concatenated to form a feature vector of $4 \times 512 = 2048$ dimensions.

This representation captures both colour distribution and coarse spatial layout, allowing better discrimination between images with similar colours but different structures. As in the baseline, the feature vector was L1-normalised.

4.3 Retrieval Performance

The spatial descriptor was evaluated using the same framework and chi-square distance metric as the baseline.

The system achieved $mAP = 0.292$, improving on the baseline result of $mAP = 0.277$. Although the improvement is modest, it shows that incorporating spatial information provides additional discriminative power.



Figure 5: Example retrieval results using the spatial RGB histogram descriptor (2×2 grid) with chi-square distance.

4.4 Effect on Retrieval Behaviour

The spatial histogram improves retrieval for categories that benefit from spatial structure, such as windows, chimneys, and cars, which tend to appear in consistent positions and exhibit characteristic patterns.

However, it does not improve all categories. Classes such as birds, signs, and miscellaneous remain challenging because the objects often occupy only a small portion of the image, while background content dominates. As a result, the descriptor is still influenced more by background colours than object-specific features.

Additionally, the 2×2 grid provides only a coarse representation of spatial layout. More advanced spatial representations could capture finer details but would increase dimensionality and computational cost.

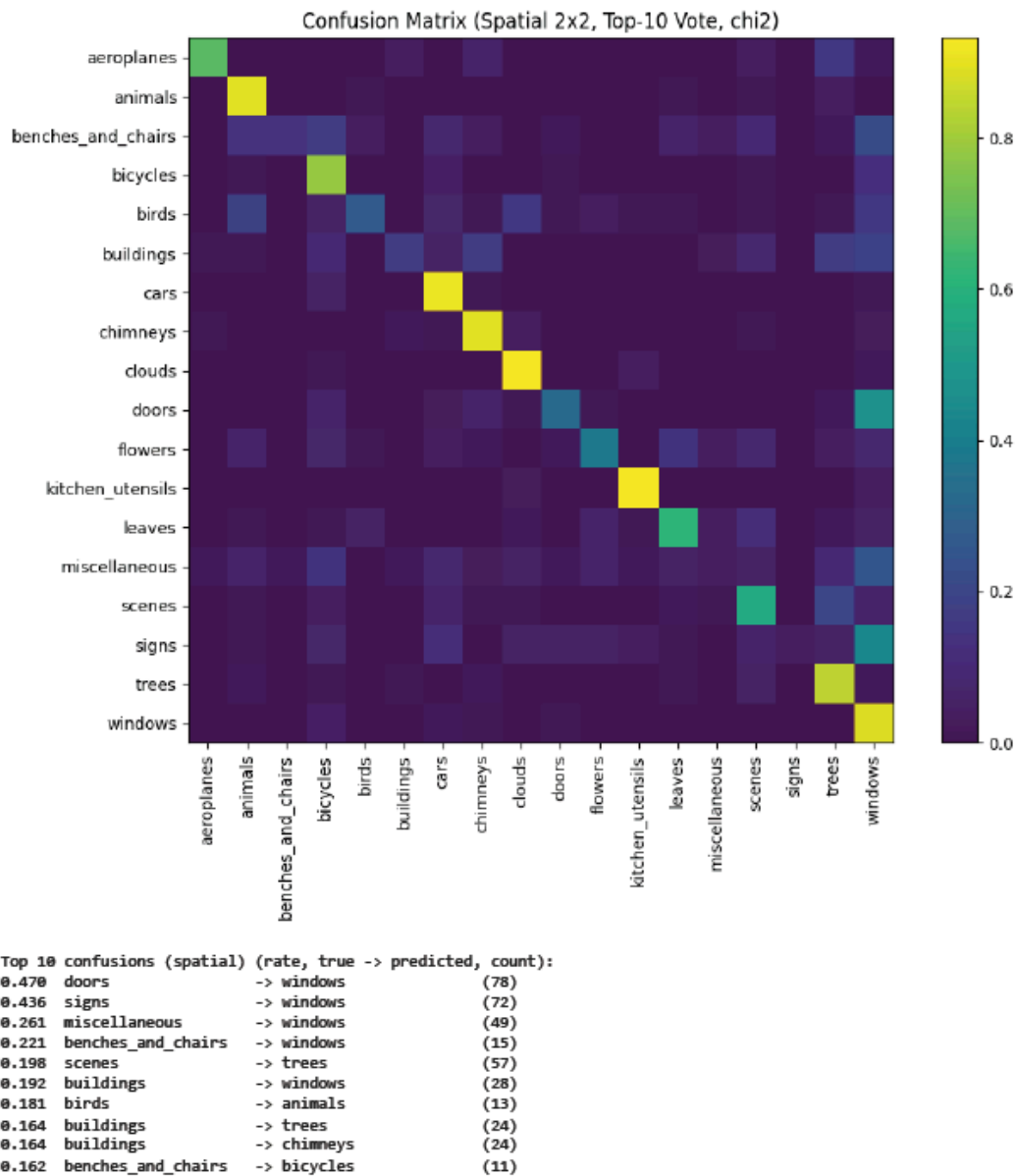


Figure 6: Confusion matrix for the spatial RGB histogram descriptor (2x2 grid) using chi-square distance.

4.5 Discussion

The spatial histogram shows that incorporating spatial structure improves retrieval performance by helping distinguish categories with similar colour distributions but different layouts.

However, it remains limited in capturing detailed object structure. More advanced descriptors that incorporate texture, shape, or local features are likely needed to achieve higher retrieval accuracy.

5. Dimensionality Reduction Using Principal Component Analysis

5.1 Motivation

The spatial histogram descriptor introduced in the previous section produces a high-dimensional feature vector of 2048 dimensions. While this representation captures both colour distribution and coarse spatial layout, such high-dimensional features may introduce redundancy and increase computational cost during retrieval.

To investigate whether a more compact representation could be obtained without significantly degrading performance, Principal Component Analysis (PCA) was applied to the spatial feature vectors. PCA is a widely used dimensionality reduction technique that projects high-dimensional data into a lower-dimensional space while preserving the directions of greatest variance in the dataset (Jolliffe and Cadima, 2016).

5.2 PCA Transformation

PCA was applied to the spatial histogram features with the goal of preserving 95% of the total variance in the data. The transformation reduced the dimensionality of the feature vectors from:

2048 $\rightarrow$ 137

dimensions. This represents a substantial reduction in feature dimensionality while retaining most of the statistical variance present in the original data.

Because PCA produces transformed features that may contain negative values and no longer represent histogram distributions, the cosine distance metric was used to measure similarity between feature vectors in the reduced feature space.

5.3 Retrieval Performance

The retrieval system was evaluated using the PCA-transformed features with cosine distance. The resulting retrieval performance was:

mAP = 0.244

This result is lower than the performance achieved with the spatial histogram descriptor before dimensionality reduction:

mAP = 0.292

5.4 Discussion

The decrease in retrieval performance indicates that reducing dimensionality through PCA can lead to a loss of discriminative information. Although PCA preserves directions of

maximum variance, these directions do not necessarily correspond to features that are most useful for distinguishing between object categories.

In addition, the original spatial descriptor represents histogram-based features, which are naturally suited to the chi-square distance metric. After applying PCA, the histogram structure is no longer preserved, making it necessary to use cosine distance instead. This change in similarity measure may also contribute to the observed reduction in performance.

Despite the decrease in mAP, PCA successfully reduces the feature dimensionality by more than an order of magnitude. This demonstrates the trade-off between computational efficiency and retrieval accuracy, which is an important consideration in large-scale image retrieval systems.

6. Discussion and Conclusion

This coursework investigated the effectiveness of several feature representations for content-based image retrieval using the MSRC image dataset. The study began with a baseline system based on a global RGB colour histogram, which captures the overall colour distribution of an image.

The baseline achieved a mean average precision (mAP) of 0.277, showing that colour information alone provides moderate retrieval performance. However, clear limitations were observed. Categories with similar colour distributions, such as doors and windows or buildings and windows, were frequently confused because global histograms ignore the spatial arrangement of visual elements.

To address this, a spatial colour histogram descriptor was introduced by dividing each image into a 2×2 grid and computing histograms for each region. This representation incorporates coarse spatial structure while retaining the advantages of colour-based features. The spatial descriptor improved performance to mAP = 0.292, indicating that spatial information helps distinguish categories with similar colours but different structures.

Principal Component Analysis (PCA) was then applied to reduce the dimensionality of the spatial descriptor from 2048 to 137 features while preserving 95% of the variance. Despite this reduction, performance decreased to mAP = 0.244, showing that PCA removes information important for retrieval. This highlights that preserving variance does not necessarily preserve discriminative features, and that PCA also disrupts the histogram structure, making distance measures such as chi-square less appropriate.

Overall, the results demonstrate that feature representation plays a critical role in image retrieval. Spatial information improves performance, while dimensionality reduction introduces a trade-off between efficiency and accuracy.

Future work could explore more advanced representations such as spatial pyramid histograms, texture descriptors, or local feature-based approaches (e.g., SIFT) (Lowe, 2004), which capture more detailed structural information. Additionally, approaches incorporating texture and layout have shown strong performance in object recognition tasks (Shotton et al., 2006).

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